Sprint Report

Unit codeEAT40006
Unit name Engineering Technology Project B (ENG/CS)-H1
Submission date: 26/08/2026
Contribution for: Sprint 4

1. ACKNOWLEDGMENT OF COUNTRY

We acknowledge the Traditional Owners of the lands on which we lived, studied, and completed this project. We pay our respects to Elders past and present and recognise their continuing connection to land, waters, and culture.

As students of Swinburne University of Technology, we acknowledge the Traditional Custodians of the land on which the university's Hawthorn campus is located, the Wurundjeri Woi-wurrung people of the Kulin Nation.

Each team member acknowledges the Traditional Owners of the land on which they resided while undertaking this work:

2. CONTRIBUTION SUMMARY (THIS SPRINT)

All team members should complete the following table together. You can provide a yes/no response or a brief comment where appropriate.

	Contribution	Initiatives	Communication	Respect
Rumeth	Contributed to literature review, simulation environment setup, system identification experiments, MPC prototype development, documentation, and sprint reporting.	Demonstrated initiative by investigating the LearningHumanoidWalking framework, conducting independent research on MPC methodologies, and proposing validation experiments.	Maintained regular communication through team meetings, supervisor meetings, email correspondence, and progress updates.	Consistently respectful, collaborative, and supportive of team discussions and decisions.
Thinidu	Contributed to literature review, simulation validation, data analysis, documentation, and project planning activities.	Demonstrated initiative by assisting with technical investigations, reviewing research findings, and supporting implementation tasks.	Actively participated in team meetings, discussions, and progress reporting activities.	Consistently respectful, cooperative, and professional when working with team members and stakeholders.

3. SPRINT PLAN

The objective of Sprint 4 was to establish the technical foundation required for implementing Whole-Body Model Predictive Control on a humanoid robot. The team focused on understanding the LearningHumanoidWalking framework, validating the Unitree H1 simulation environment, analysing robot dynamics, and developing an initial MPC prototype.

Report: MPC Sprint 1

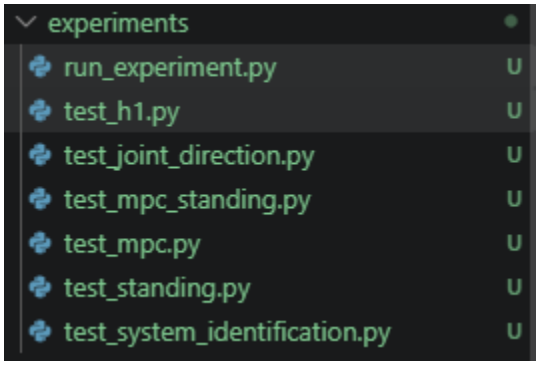
Scope changes log

Date ↑	Key	Summary	Work type
2026-08-12	MPC-1	Improve Balance Foundation	Story
2026-08-12	MPC-2	Advanced Balance Control	Story
2026-08-12	MPC-3	Recovery Stepping	Story
2026-08-12	MPC-4	Sprint Integration & Testing	Story
2026-08-26	MPC-3	Recovery Stepping	Story
2026-08-26	MPC-4	Sprint Integration & Testing	Story
2026-08-26	MPC-1	Improve Balance Foundation	Story
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2026-08-26	MPC-3	Recovery Stepping	Story
2026-08-26	MPC-4	Sprint Integration & Testing	Story
2026-08-26	MPC-2	Advanced Balance Control	Story

Sprint Backlog

Task	Priority	Status
Configure simulation environment	High	Completed
Validate H1 standing environment	High	Completed
Validate H1 walking environment	High	Completed
Analyse joint behaviour	High	Completed
Conduct system identification	High	Completed
Implement reduced-order MPC	High	Completed
Evaluate controller performance	Medium	Completed

Artifacts Produced



4. RISK MANAGEMENT PLAN

Updated Risk Register

Risk	Probability	Impact	Mitigation
Client communication delays	Medium	High	Schedule regular meetings
MPC implementation complexity	High	High	Begin with simplified models
MuJoCo simulation issues	Medium	High	Validate components individually
Lack of real robot access	High	Medium	Use simulation-based validation
Team member availability	Medium	Medium	Weekly planning meetings
Project scope expansion	Medium	High	Focus on standing and walking control

5. SPRINT PROGRESS

a) Standing Control Analysis

The standing environment was tested to understand the baseline behaviour of the robot under existing control policies. The objective was to verify that the robot could maintain an upright posture and to identify the conditions that caused instability or episode termination.

Experimental results showed that the robot was capable of maintaining balance for a short period before the simulation terminated. Termination occurred when the robot's pelvis height dropped below the minimum threshold defined by the environment. This behaviour confirmed that the safety constraints implemented within the framework were functioning correctly.

The standing tests provided valuable insight into the balance characteristics of the robot and established a baseline against which future MPC controllers can be compared

Results showed:

- Robot maintained balance briefly.
- Termination occurred due to height reduction below threshold.
- Baseline controller behaviour successfully understood.

```
(learninghumanoidwalking) PS C:\Users\rumet\Desktop\MPC test 2\LearningHumanoidWalking\experiments> python test_h1.py
H1 environment started
Observation shape: (43,)
Step:   0 | Height: 0.983 | Collision: False | Reward: 0.204
Step:   1 | Height: 0.977 | Collision: False | Reward: 0.075
Step:   2 | Height: 0.965 | Collision: False | Reward: 0.439
Step:   3 | Height: 0.949 | Collision: False | Reward: 0.518
Step:   4 | Height: 0.929 | Collision: False | Reward: 0.701
Step:   5 | Height: 0.907 | Collision: False | Reward: 0.718
Step:   6 | Height: 0.882 | Collision: False | Reward: 0.486
Step:   7 | Height: 0.857 | Collision: False | Reward: 0.475
Step:   8 | Height: 0.830 | Collision: False | Reward: 0.422
Step:   9 | Height: 0.803 | Collision: False | Reward: 0.396
Step:  10 | Height: 0.775 | Collision: False | Reward: 0.388
Step:  11 | Height: 0.747 | Collision: False | Reward: 0.384
Step:  12 | Height: 0.719 | Collision: False | Reward: 0.360
Step:  13 | Height: 0.691 | Collision: False | Reward: 0.354
Step:  14 | Height: 0.664 | Collision: False | Reward: 0.347
Step:  15 | Height: 0.637 | Collision: False | Reward: 0.342
Step:  16 | Height: 0.611 | Collision: False | Reward: 0.340
Step:  17 | Height: 0.586 | Collision: False | Reward: 0.436

========== TERMINATED ==========
REASON: H1 fell below 0.6 m
================================
Finished
```

```
(learninghumanoidwalking) PS C:\Users\rumet\Desktop\MPC test 2\LearningHumanoidWalking\experiments> python test_mpc_standing.py

==========================================
H1 FEEDBACK STANDING EXPERIMENT
==========================================
Environment created
Observation shape: (35,)
Step:   0 | Height: 0.994 | Roll:   2.38° | Pitch:   0.92° | PitchVel:  -0.032 | Reward: 0.995

==========================================
CONTROLLER TERMINATED
==========================================
Step:   6
Height: 0.935
Pitch:  -42.81°
Total reward: 4.554

Finished.
```

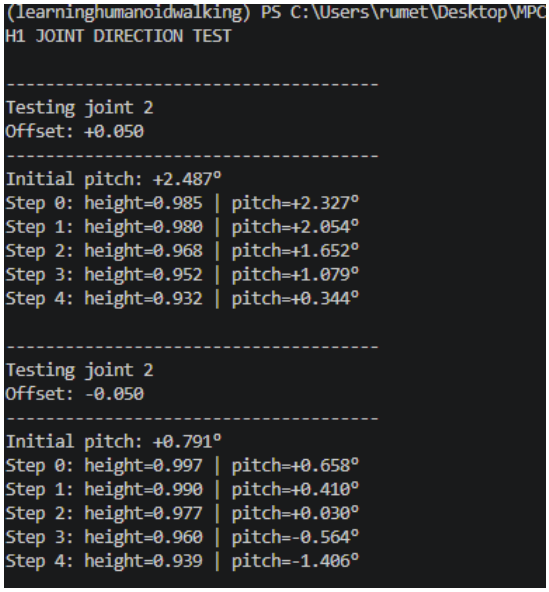
b) Joint Direction Testing

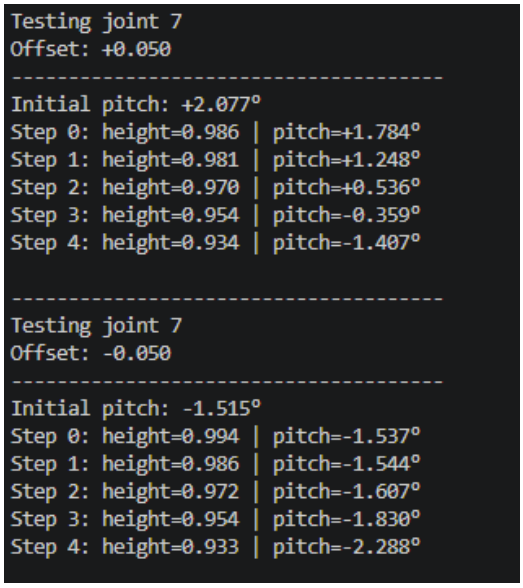
A detailed investigation was conducted to determine how individual joints contribute to maintaining balance and posture. Controlled perturbations were applied to selected joints while observing changes in robot orientation and body position.

The analysis focused on the left hip, left knee, right hip, and right knee joints because these joints play a significant role in sagittal-plane stability. By applying positive and negative joint movements, the team observed corresponding changes in pelvis orientation and body pitch angle.

Results showed that hip joints primarily influenced upper-body orientation, while knee joints significantly affected pelvis height and overall centre-of-mass positioning. Understanding these relationships is essential for MPC development because the controller must determine which joints should be adjusted to restore balance following disturbances.

The experiments also identified the most influential joints for balance recovery, providing valuable information for selecting control inputs in future reduced-order and whole-body models.





Experiments were conducted on:

- Left Hip
- Left Knee
- Right Hip
- Right Knee

Results identified:

- Joint influence on pitch dynamics.
- Joint response directions.
- Key control joints for balance recovery.

c) System Identification

System identification experiments were performed to obtain an initial understanding of the dynamic behaviour of the humanoid robot. Small control perturbations were applied while monitoring the resulting motion of the robot.

```
(learninghumanoidwalking) PS C:\Users\rumet\Desktop\MPC test 2\LearningHumanoidWalking\experiments> python test_system_identification.py

============================================
H1 SYSTEM IDENTIFICATION
============================================

--------------------------------------------
LEFT_HIP | joint=2 | offset=-0.050
--------------------------------------------
 0 | h=0.9833 | pitch=-2.027° | pitch_vel=-0.06236
 1 | h=0.9824 | pitch=-2.189° | pitch_vel=-0.17458
 2 | h=0.9768 | pitch=-2.399° | pitch_vel=-0.15371
 3 | h=0.9654 | pitch=-2.677° | pitch_vel=-0.23709
 4 | h=0.9494 | pitch=-3.080° | pitch_vel=-0.31680
 5 | h=0.9296 | pitch=-3.609° | pitch_vel=-0.47145
 6 | h=0.9076 | pitch=-4.537° | pitch_vel=-0.80418

--------------------------------------------
LEFT_HIP | joint=2 | offset=+0.050
--------------------------------------------
 0 | h=0.9929 | pitch=+1.306° | pitch_vel=-0.03064
 1 | h=0.9908 | pitch=+1.232° | pitch_vel=-0.10598
 2 | h=0.9834 | pitch=+1.037° | pitch_vel=-0.19836
 3 | h=0.9705 | pitch=+0.635° | pitch_vel=-0.35734
 4 | h=0.9527 | pitch=+0.012° | pitch_vel=-0.50858
 5 | h=0.9313 | pitch=-0.828° | pitch_vel=-0.68911
 6 | h=0.9073 | pitch=-1.984° | pitch_vel=-0.91726

--------------------------------------------
```

Three key state variables were recorded:

- Pelvis height
- Pitch angle
- Pitch angular velocity

The collected data allowed the team to observe how the robot responded to external disturbances and control actions over time. Analysis of these responses provided insight into the stability characteristics of the system and revealed the relationship between joint inputs and body motion.

These findings are important because MPC relies on predictive models of system dynamics. The system identification process therefore represents an essential step towards constructing more accurate predictive control models in future sprints.

d) Reduced-Order MPC Prototype

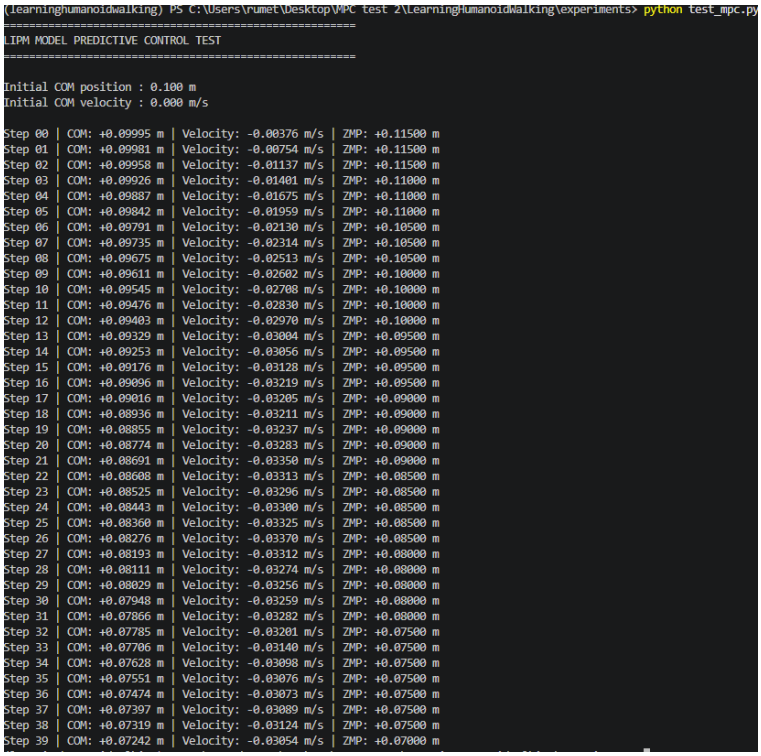
A reduced-order Model Predictive Controller based on the Linear Inverted Pendulum Model (LIPM) was implemented as an initial proof of concept.

The LIPM simplifies the humanoid robot into a point mass located at the centre of mass (COM) supported by a massless leg. Although simplified, this model captures the fundamental balance dynamics required for preliminary MPC development.

The controller was designed to minimise centre-of-mass tracking error while maintaining Zero Moment Point (ZMP) stability constraints. At each control step, the MPC solved an optimisation problem to determine the optimal future control actions over a prediction horizon.

Experimental results demonstrated successful regulation of the COM position. The initial COM displacement of 0.100 m was reduced to approximately 0.072 m after 40 control steps.

This result confirms that predictive control can gradually stabilise the system while satisfying balance constraints. The successful implementation of the reduced-order controller provides confidence that MPC techniques can be extended to more complex humanoid models in future stages of the project



6. SPRINT REVIEW

a) Critical Analysis of Sprint Progress

A transition from environment investigation to controller development is represented in sprint 4

The most significant achievement was not yet stable humanoid locomotion, but the establishment of the technical components required to achieve it. The simulation environment is operational, robot states can be observed, joint responses have been experimentally investigated, and an independent reduced-order MPC model has been implemented and validated.

One important finding was that directly applying feedback corrections to humanoid joints without sufficiently understanding the robot dynamics can rapidly destabilise the system. Early

feedback standing controller resulted in excessive torso pitch and termination of the experiment. Rather than continuing to tune gains without a model, the team changed approach and conducted joint influence and system identification experiments.

This represents an important improvement in the development methodology. Future controller integration will therefore be based on experimentally observed robot behaviour and a structured reduced-order model.

The primary limitation at the end of Sprint 4 is that the LIPM MPC and the H1 simulation currently operate as validated components rather than as a fully integrated controller. Integration of COM state estimation, MPC output and humanoid balance control is therefore the primary objective of Sprint 5.

b) Professional Conduct and Communication

Our communication was handled professionally throughout the sprint through regular communication between team members and preparation of technical progress for client meetings.

Other than hiding the technical problems were shown including unsuccessful standing controller experiments. This allowed the team to explain both successful results and limitations during project discussions.

The presentation was structured around measurable evidence including simulation outputs, pelvis height, torso orientation, joint responses, COM trajectories and MPC generated control commands.

This approach allows project progress to be evaluated objectively rather than relying solely on visual simulation results.

7. CLIENT FEEDBACK

8. RETROSPECT (CRITICAL REVIEW OF THE PROCESS)

a) Process Review

A considerable amount of technical investigation was involved because the project moved from theoretical research towards practical implementation on a high dimensional humanoid simulation.

Several sides of the development process worked effectively.

We did regular communication and divided technical tasks according to the immediate requirements of the project. Experimental results and simulation behavior were shared between members, which allowed us to take decisions based on technical observations.

Another strength was the incremental development strategy. Other than attempting to implement the complete Whole Body MPC architecture immediately, the problem was divided into smaller components:

1. Validate the H1 simulation environment.
2. Validate standing and walking environments.
3. Investigate baseline stability.
4. Identify joint influence on body dynamics.
5. Perform preliminary system identification.
6. Develop and validate a reduced-order LIPM model.
7. Implement an initial MPC controller.

This minimizes the difficulty of identifying errors because each component could be tested independently.

b) Challenges Encountered

One of the key challenges during sprint 4 was the complexity of humanoid dynamics.

Unlike simpler robotic systems, humanoid balance depends on interactions between several joints, contact forces, Centre of Mass motion, torso orientation and ground contact. A control input intended to correct torso pitch may simultaneously affect pelvis height or another aspect of the robot's stability.

The second biggest challenge was understanding the existing H1 simulation repository. The framework contains multiple abstractions for environments, tasks, robot interfaces, reward functions, and PD control. Considerable investigation was therefore required before modifications could be made safely.

Dependency and model configuration also created primary implementation difficulties. The H1 environment required the correct MuJoCo Menagerie model files before the simulation could be executed successfully.

Initial controller feedback experiment was one of the other challenges we face. Direct torso pitch corrections caused the robot to become unstable rapidly. This shows that the relationship between high-level balance errors and joint level actions was more complex than initially assumed.

by introducing dedicated joint direction and system identification experiments rather than continuing uncontrolled gain tuning.

c) Improvements to the Development Process

The sprint showed that controller development should continue using an incremental experimental approach.

Future changes will first be tested in isolated scripts before being integrated into the main controller. Controller parameters, simulation states, and experimental results will also be logged to allow for quantitative comparison between controller versions.

The development process will therefore follow the sequence:

model → validate → integrate → test → evaluate

rather than attempting simultaneous development of all controller components.

This approach is expected to reduce debugging time during Sprint 5 and Sprint 6.

d) Digital Ethics and Security Review

The project currently operates entirely within a simulated MuJoCo environment and does not collect, process or store personal information. Consequently, there are currently no direct privacy concerns associated with implementation.

However, ethical considerations remain relevant because the project concerns humanoid robots intended for potential industrial applications.

A malfunctioning control system can create significant safety risks if transferred from simulation to a physical humanoid robot. Therefore, controller stability, predictable behaviour, force limitations, and failure detection are important considerations when evaluating the proposed system.

The current project reduces these risks by conducting controller development and experimentation exclusively in simulation. Potentially unstable controllers can therefore be evaluated without exposing humans or physical equipment to risk.

No external credentials, personal datasets, or confidential client information are required by the current controller implementation. Project code and external repositories should nevertheless continue to be managed according to appropriate software development practices, including source attribution and version control.

e) Team Code of Conduct Review

We reviewed its agreed working practices during Sprint 4 and found that the project continued to operate consistently with the established code of conduct.

We maintained regular communication, shared technical progress, attended scheduled discussions where possible, and supported each other when implementation problems occurred.

Other than individual preference, technical disagreements were addressed through experimentation and evidence, for example, unsuccessful controller behaviour was investigated using additional experiments before deciding on the next implementation approach.

No significant team conduct issues were identified during this sprint.

We will continue to maintain clear task ownership and regular progress communication during Sprint 5 because controller integration will require close coordination between simulation, modelling, and MPC development.

9. LESSONS LEARNED (CRITICAL REVIEW OF SPRINT ONE EXPERIENCE AND FUTURE PLAN)

a) 9.1 Lessons Learned

- Sprint 4 provided several important technical and project management lessons.
- incremental development approach is required by humanoid control
- Trying to control the complete humanoid system immediately creates significant debugging difficulty because failures can originate from joint mappings, state estimation, controller gains, contact dynamics or the predictive model. Breaking the problem into independently testable components provides a more reliable development process.
- A stable nominal pose is not automatically producing stable dynamic behaviour**.**
- The experiments demonstrated that commanding the H1 robot to remain at its predefined half sitting pose does not guarantee balance. Although PD control attempts to maintain joint positions, the robot still requires active balance corrections to maintain its Centre of Mass relative to the support region.
- before controller integration system modification is important
- The joint influence and system identification experiments demonstrated that individual joint changes can significantly affect torso orientation and pelvis height. Understanding these relationships is important when converting MPC outputs into joint-level commands.
- Reduced order models provide a practical starting point for humanoid MPC.
- Direct optimisation over the complete nonlinear dynamics of a humanoid robot would significantly increase computational and implementation complexity. The LIPM provides a simpler representation of horizontal COM dynamics and allows the fundamental MPC architecture to be developed and validated before increasing controller complexity.

- useful engineering evidence is provided by unsuccessful experiments
- The initial feedback standing controller did not stabilise the humanoid. somehow, the failure identified an important limitation in the original control strategy and motivated the subsequent system identification experiments. Recording unsuccessful experiments therefore contributed directly to the controller design process.

b) Sprint 5 – MPC Integration and Balance Control

- The main objective of Sprint 5 will be to integrate the reduced order MPC controller with the H1 humanoid simulation.
  The planned work includes
- Extract the H1 Centre of Mass position directly from MuJoCo.
- Calculate or estimate COM velocity.
- Determine the support region using the robot's foot positions.
- Generate feasible ZMP reference commands.
- Provide the measured COM state to the MPC at each control interval.
- Generate MPC predictions over a finite horizon.
- Convert MPC balance commands into appropriate lower-body joint targets.
- Integrate the MPC layer with the existing H1 joint-level PD controller.
- Perform standing-balance experiments.
- Compare baseline standing behaviour against MPC-assisted standing.
- Record quantitative metrics including survival time, COM displacement, torso pitch and pelvis height.

c) Sprint 6 – Walking, Evaluation and Final Demonstration

- Sprint 6 will mainly focus on extending the controller from stationary balance towards locomotion and completing the final project evaluation.
- The planned activities include:
- Extend the MPC framework to support changing support conditions during walking.
- Generate COM/ZMP references compatible with a basic walking sequence.
- Implementing predictive balance corrections with the H1 walking behaviour.
- Investigate target directed forward locomotion where feasible.
- Introduce controlled disturbances to evaluate controller robustness.
- Compare controlled and baseline experiments.
- Measure stability, COM tracking, torso orientation and episode duration.
- Record simulation demonstrations for the final presentation.
- Complete technical documentation and final project evaluation.

d) Overall Reflection

- Sprint 4 shows the transition from research and simulation setup towards implementation of the project's main predictive control methodology.
- Even the complete Whole Body MPC controller has not yet been achieved, the project now contains the essential foundations required for further development:
- The major lesson from this sprint is that successful humanoid control depends on systematic integration other than immediately increasing controller complexity. Sprint 5 will therefore prioritise reliable MPC based standing balance before Sprint 6 extends the system towards locomotion and final evaluation.
- This planned approach provides a practical path towards completing a defensible prototype of the elected Whole Body Model Predictive Control for Industrial Humanoids system by the end of the project.